

1. True or false: When an OSPF router sends its link information, it is sent only to those nodes directly attached neighbors. Explain the reason.

ANS: False

Link information can be forwarded through another router, so the node not directly connected still can received the messages.

2. Will a BGP router always choose the loop-free route with the shortest AS path length? State your reason.

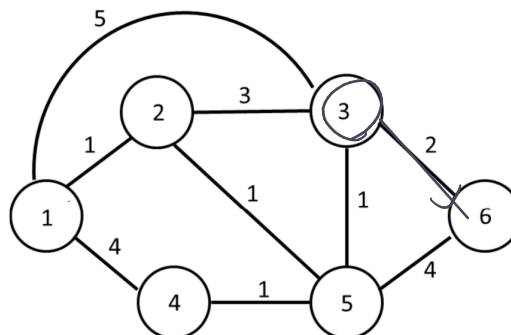
ANS: Not always

Most of the BGP routers are policy based, so it will first choose the route by its policy prior to the shortest path

3. It is preferable to send SNMP messages in unreliable UDP datagram. Why do you think the designers of SNMP chose UDP rather than TCP as the transport protocol of choice for SNMP?

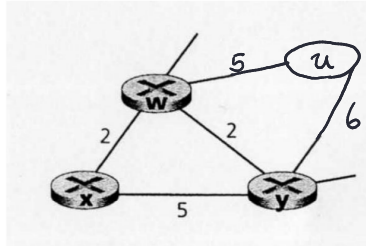
ANS: Because UDP can deliver message under bad network condition, and its low-load characteristic made it be the best choice for SNMP

4. Use the Dijkstra's link-state algorithm to find the least-cost paths from Node 1 to all other nodes of the following graph. The number above each link is the associated cost of the link. **Show your results step by step.**



| Step | N'               | D(2), p(2) | D(3), p(3) | D(4), p(4) | D(5), p(5) | D(6), p(6) |
|------|------------------|------------|------------|------------|------------|------------|
| 0    | 1                | (1, 1)     | (5, 1)     | (4, 1)     | $\infty$   | $\infty$   |
| 1    | 1, 2             | x          | (4, 2)     | (4, 1)     | (2, 2)     | $\infty$   |
| 2    | 1, 2, 5          | x          | (3, 5)     | (3, 5)     | x          | (6, 5)     |
| 3    | 1, 2, 5, 4       | x          | (3, 5)     | x          | x          | (6, 5)     |
| 4    | 1, 2, 5, 4, 3    | x          | x          | x          | x          | (5, 3)     |
| 5    | 1, 2, 5, 4, 3, 6 | x          | x          | x          | x          | x          |

5. Consider the network fragment shown below.  $x$  has only two attached neighbors,  $w$  and  $y$ .  $w$  has a minimum-cost path to destination  $u$  (not shown) of 5, and  $y$  has a minimum-cost path to  $u$  of 6. The complete paths from  $w$  and  $y$  to  $u$  (and between  $w$  and  $y$ ) are not shown. Assume all link costs have strictly positive integer values.



- a. Give  $x$ 's distance vector for destinations  $w$ ,  $y$  and  $u$ .

$$DV_x(w) = \min \{C_{x,v} + D_v(w)\} = 2 + 0 = 2 \#$$

$$\hookrightarrow D_y(w) : \min \{C_{y,v} + D_v(w)\} = 2$$

$$\hookrightarrow D_w(w) : 0 \checkmark$$

$$DV_x(y) = \min \{C_{x,v} + D_v(y)\} = 2 + 2 = 4 \#$$

$$\hookrightarrow D_y(y) : 0 \checkmark$$

$$\hookrightarrow D_w(y) : \min \{C_{w,v} + D_v(y)\} = 2$$

$$DV_x(u) = \min \{C_{x,v} + D_v(u)\} = 2 + 5 = 7 \#$$

$$\hookrightarrow D_y(u) : \min \{C_{y,v} + D_v(u)\} = 6$$

$$\hookrightarrow D_w(u) : \min \{C_{w,v} + D_v(y)\} = 5$$

- b. Discuss the condition where a link-cost change for either  $c(x,w)$  or  $c(x,y)$  will trigger  $x$  to inform its neighbors of a new minimum-cost path to  $u$  after executing the distance-vector algorithm.

ANS: (1) Any change in  $c(x,w)$  will make  $x$  inform its neighbor

of a new minimum-cost to  $u$ , because  $c(x, w)$  is a part of the current minimal path

(2) Any change in  $c(x,y)$  won't make  $x$  inform its neighbor,

since the path to  $u$  including is  $c(x,y) + c(y,u)$  which will be at least 7, unless  $c(x,y)$  is less than 1, or the path couldn't be any shorter. So any change of  $c(x,y)$  won't affect the minimal path.